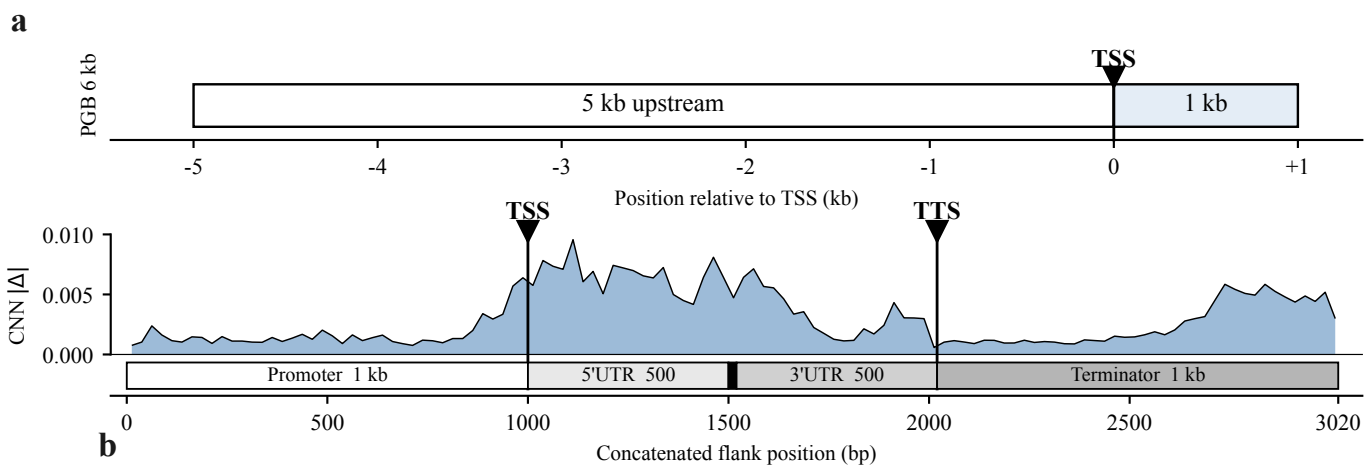


b

PGB Arabidopsis sequence windows

6 kb = 5 kb upstream + 1 kb downstream of TSS
used for training and primary AUROC

56-tissue / sample abundances

gene-level matrix paired to the same genes
56 tissues / samples per gene

Official PGB gene split, no overlap

Train 25,731 · Valid. 3,401 · Test 3,402
32,534 genes after the public partition

→ Public split kept as published

Tissue-averaged binary labels

$\bar{y}_g$ over 56 tissues; $z_g = \mathbf{1}\{\bar{y}_g \geq \tau\}$
$\tau =$ train-set median $= 5.605$

→ Derived binary task; not the official PGB multi-tissue regression

1D-CNN · DeepSEA-style

One-hot 6 kb × 4
3 conv. blocks; k=8, 256; pool 4
global max-pool + FC
2.7 M · 30 epochs
AdamW 1×10^{-3}, 15 min
seeds 42–46

AgroNT-1B · LoRA

6-mer tokens; frozen body
LoRA on Q/K/V/O
$W = W_0 + (\alpha/r)BA$; $r = 16$, $\alpha = 32$
16.2 M of 985 M (1.6%)
AdamW 2×10^{-4}, 3 epochs, 240 min
seeds 42–46

→ Identical splits, labels, and metrics one RTX 4090

Four locked readouts

1. Arabidopsis test n=3,402 — AUROC 0.842 vs 0.939; McNemar 532 vs 142
2. Region maps — exploratory; allele protocols differ by architecture
3. Natural SNVs — 250 genes; CNN 27,662 sites; AgroNT 26,518 sites
4. Crop PGB transfer — rice, maize, tomato, soybean; AgroNT saw those genomes